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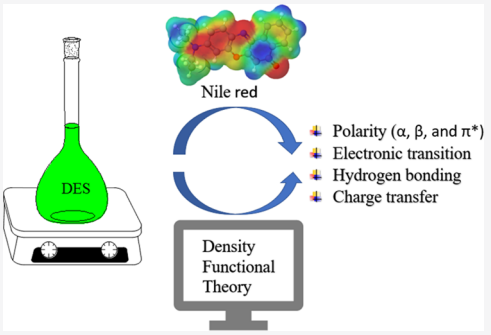
Solvatochromic Parameters of Deep Eutectic Solvents: Effect of Different Carboxylic Acids as Hydrogen Bond Donor

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Cite This: J. Chem. Eng. Data 2020, 65, 640−646 Read Online

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ABSTRACT: The physicochemical properties of carboxylic acid-based deep eutectic solvents (DESs) were measured by solvatochromic probes: Nile red, *N*,*N*-diethyl-4-nitroaniline, and 4-nitroaniline. UV−vis absorbance of these probes was computed with temperature in the range of 298.15 to 318.15 K. By increasing temperature, the normalized polarity of DESs stayed linear except for [Ch]Cl:butyric acid (1:3) and [Ch]Cl:caprylic acid (1:2), which had a modest shift in the maximum wavelength between T = (308.15 to 318.15) K. Changes in the normalized polarity parameter of DESs demonstrated an interesting trend, and it confirms that alkyl chain length of the hydrogen bond donor (HBD) affects the solvation microsphere of the solute−solvent interaction. Also, dipolarity/polarizability (π^*) and hydrogen bond acidity (α) decrease with increasing alkyl chain length confirming the significant contribution of HBDs to Kamlet−Taft parameters. Interestingly, the π^* and α parameters of DESs did not change significantly and seem to be independent of temperature. In addition, density functional theory was used to extend our understanding on how alkyl chain length of HBD affects charge transfer in the DESs. In summary, both solvatochromic molecular probes and density functional theory are effective methods for studying polarity parameters and charge transfer, respectively, for DESs with varying alkyl chain length of the hydrogen bond donor.



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1. INTRODUCTION

Deep eutectic solvents (DESs) are a new generation of designer solvents alternative to ionic liquids.[1,2] The term was first introduced by Abbott and his co-workers by mixing choline chloride and urea at 1:2 molar ratio, with a depressed melting point compared to the individual components.[1,2] Simply put, the formation of DESs involves the combination of a salt acting as a hydrogen bond acceptor (HBA) with a hydrogen bond donor (HBD) in the correct molar ratio until the eutectic point is reached. The driving force for the eutectic point is hydrogen bonding interaction between the chloride anion of the salt and the proton of the HBD, resulting in charge delocalization.[2,3] Since its first inception, many independent researchers have worked with different HBA and HBD to understand the nature, physicochemical behavior, and thermophysical properties of the solvent.[4−7] Choline chloride, [Ch]Cl, remains the most used salt for DES formation due to its low cost, low toxicity, biodegradability, and biocompatibility whereas some commonly used HBDs include urea, alcohol, sugar, amide, amino acid, and carboxylic acid.[7,8]

Carboxylic acid-based DESs have recently attracted attention due to their uniqueness in analytical extraction,[9,10] organic synthesis,[11] electrochemistry,[6] and pharmaceutics.[12] Commonly used monocarboxylic acids, dicarboxylic acids, and tricarboxylic acids in DES formulation include acetic acid, oxalic acid, malonic acid, malic acid, maleic acid, levulinic acid, and citric acid.[13,14] Recently, a hydrophobic DES was synthesized from decanoic acid and quaternary ammonium salts and the solvent was found to have higher extraction efficiency for volatile organic compounds (VOCs).[15]

Despite the numerous physicochemical and thermochemical data on carboxylic acid-based DESs, little or no attempt has been made to understand the molecular interactions that occur within these solvent systems. For example, a pioneering work by Abbott et al.[1,2] inferred that two carboxylic acids might be required to complex each chloride anion for the eutectic point to be reached. Over a decade of this postulation, a detailed molecular explanation on how this complexation occurs is still missing in the literature. Recently, the solvatochromic properties and molecular dynamic simulations were performed on DESs formulated from choline chloride and urea, ethaline, and glycerol to show hydrogen bonding.[16]

In this work, DESs with different molar ratios of choline chloride and butyric acid (BA), valeric acid (VA), and caprylic acid (CA) were studied by using solvatochromic probes and density functional theory (DFT). For DFT, the frontier molecular orbitals (FMOs), which are known to contain useful information for chemical reactions, were used to study charge transfer of DES mixtures.[17−20] Moreover, these hydrogen bond

Received: September 8, 2019
Accepted: December 19, 2019
Published: January 6, 2020

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https://dx.doi.org/10.1021/acs.jced.9b00872
J. Chem. Eng. Data 2020, 65, 640−646

donors were selected not only because they have not been thoroughly investigated in the literature but also to monitor how a change in their alkyl chain length may be used as an innovative tool to tailor DES polarity. To date, most DESs have been hydrophilic,[21] hence studying the solvatochromic and charge transfer properties of these hydrophobic DESs would be the first of its kind. The longest wavelength intermolecular charge transfer polarity scale by Kosower[22] was normalized by Nile red (NR) dye to determine the electron transition (E_T or Z) (eq 1).[23] The Kamlet−Taft (KT) multiparameter linear polarity parameters (hydrogen bonding acidity, α; hydrogen bonding basicity, β; dipolarity/polarizability, π^*) were determined using 4-nitroaniline (4NA) and *N*,*N*-diethyl-4-nitroaniline (DENA) as the molecular solvatochromic probes (eqs 2−4).

To fully understand how hydrogen bond donors alter charge transfer, dipole moments, and other thermophysical properties of DESs, a density functional theory (DFT) employing Beck's three-parameter exchange functional[24] and Lee−Yang−Parr correlation functional[25] (B3LYP) at the 6-312 (d) basis set was used to study the FMOs. In this study, experimental procedures by using solvatochromic molecular probes and DFT were explored to help understand how hydrogen bond donors and hydrogen bond acceptors behave during DES formulation.

2. EXPERIMENTAL SECTION

2.1. Chemicals. Choline chloride, butyric acid, caprylic acid, Nile red, and 4-nitroaniline were all purchased from Acros Organics. Valeric acid was purchased from Alfa Aesar and *N*,*N*-diethyl-4-nitroaniline was from Frinton Laboratories, Inc. Table 1 lists the purities, suppliers, CAS numbers, and purification methods for all chemicals used. All other chemicals were of analytical grade.

Table 1. Chemical Sample Information

chemical name	CAS reg. no.	suppliers	mass fraction purity	purification method
choline chloride	67-48-1	ACROS Organics	0.990	TA[a]
butyric acid	107-92-6	ACROS Organics	≥0.990	GC[b]
valeric acid	109-52-4	Alfa Aesar	0.990	GC
caprylic acid	124-07-2	ACROS Organics	0.990	GC
Nile red	7385-67-3	ACROS Organics	0.990	UV[c]
4-nitroaniline	100-01-6	ACROS Organics	≥0.990	GC
N,*N*-diethyl-4-nitroaniline	2216-15-1	Frinton Laboratories	0.980	titration/GC

[a]Titration argentometric. [b]Gas chromatograph. [c]Ultraviolet spectrophotometry.

2.2. Synthesis of DESs. DESs were synthesized by following the procedure described[14] with some modifications. This involves mixing choline chloride and butyric acid, valeric acid, and caprylic acid at 1:2 and 1:3 molar ratios. The mixture was heated at 80 °C for 15−30 min under constant stirring at an atmospheric pressure until a homogenous liquid was obtained. The prepared DESs were cooled to room temperature and stored in a desiccator before use.

2.3. Solvatochromism. A UV−vis spectrophotometer (Synergy H1 microplate reader, Biotek) was used to determine the maximum absorption wavelength (λ_{max}) of DESs in solvatochromic probes. Absorbance readings were collected by spectral scanning of DESs in Nile red at λ = (400 to 650) nm, *N*,*N*-diethyl-4-nitroaniline at λ = (300 to 450) nm, and 4-nitroaniline at λ = (300 to 450) nm at 1 nm interval in triplicate. The working concentration for all the solvatochromic probes was kept as low as 0.5×10^{-4} M to help neglect the effect of solute−solute interaction.[26] The effect of temperature on electron transition in the probes and on the Kamlet−Taft parameters was obtained at T = (298.15 to 318.15) K. The unit of electron transition in Nile red, E_T (NR), is in kcal·mol^{-1}, υ is the wavenumber, and λ_{max} is the maximum wavelength in nanometer (nm). The calculations of E_T (NR), α, β, and π^* values use the equations[27−29]

$$E_T\ (\mathrm{NR})\ \mathrm{kcal{\cdot}mol^{-1}} = 28591.44/\lambda_{max}\ (\mathrm{nm}) \quad (1)$$

$$\alpha = (19.9657 - 1.0241\pi^* - \upsilon_{NR})/1.6078 \quad (2)$$

$$\beta = (1.035\upsilon_{DENA} + 2.64 - \upsilon_{4NA})/2.80 \quad (3)$$

$$\pi^* = 0.314\ (27.52 - \upsilon_{DENA}) \quad (4)$$

where $\upsilon = 1/(\lambda_{max} \times 10^{-4})$.

2.4. Computational Method. Quantum theoretical calculations were performed with the Gaussian 09 software package integrated with the Gauss View 5.0.8 graphical interface.[30] Gas-phase equilibrium geometries were fully optimized with vibrational frequencies calculated using DFT/B3LYP at the 6-31G (d) basis set. The DFT/B3LYP functional makes correction for both gradient and exchange correlations and provides qualitative results, which have comparable accuracy compared to ab initio methods at a reduced CPU time. The absence of imaginary frequency showed that structures are at true minima.[31] The molecular orbital calculations for optimized HBA, HBD, and DESs were executed with the same computational method and basis sets at 298 K to obtain the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) energies. The HOMO−LUMO gap and charge transfer (ΔN) of DESs were calculated.

3. RESULTS AND DISCUSSION

3.1. Physical Characterization of DESs. Several carboxylic acid-based DESs have been reported in the literature, but data on the electronic properties of these solvents are sporadic. Also, little or no work has been done to elucidate how an increased hydrocarbon chain length of fatty acids affects solvatochromic parameters and charge transfer of DESs. In this work, three hydrogen bond donors (HBDs) were selected to tailor the physicochemical behavior of DESs by increasing their alkyl chain length. Butyric acid (BA), valeric acid (VA), and caprylic acid (CA) are cheap, readily available, biodegradable, and nontoxic making them excellent HBDs for this research. In Table 2, choline chloride with either butyric acid or valeric acid at 1:3 molar ratio forms a transparent liquid at room temperature, whereas choline chloride with caprylic acid at the same molar ratio appears cloudy. The high viscosities of [Ch]Cl:BA, [Ch]Cl:VA, and [Ch]Cl:CA DESs (1:2) are partly due to hydrogen bonding and charge transfer occurring between the cation and the chloride anion (Table S1). DESs at 1:1 molar ratio stay solid at room temperature.

https://dx.doi.org/10.1021/acs.jced.9b00872
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Table 2. Physical Characterization of Formulated DESs at 298 KM[a,b]

DESs	acronym	physical characterization	
		molar ratio 1:2	molar ratio 1:3
choline chloride:-butyric acid	[Ch]Cl:BA	viscous (high), white cloudy	white transparent liquid, viscous (medium)
choline chloride:-valeric acid	[Ch]Cl:VA	viscous (high), white cloudy	white transparent liquid, viscous (medium)
choline chloride:-caprylic acid	[Ch]Cl:CA	viscous (very high), white yellowish	viscous (very high), white yellowish

[a]Calculated viscosities for DESs are listed in Table S1. [b]Standard uncertainty, u, of T is $u(T)$ = 0.2 K.

3.2. Solvatochromism. *3.2.1. Electron Transition (E_T) between Solvent and Solute.* Solvatochromism, due to its simplicity, remains the most widely used method for studying empirical solvent polarity because the spectrum of the probes (solute) changes when it dissolves in different solvent molecules in the microsphere solvation of probes. The use of a 96-well microplate UV−vis spectrophotometer equipped with a 4-Zone™ incubator made it easier to control temperature between T = (298 and 318) K. Nile red, instead of penta-*tert*-butyl-substituted betaine dye by Dimroth and Reichardt, was used to calculate E_T values due to the highly acidic character of the DESs. The latter becomes protonated in acidic medium and looses its absorption band, thus rendering it an inappropriate probe for the solute−solvent interaction. The low basicity (pK_a = 1.00 ± 0.05), photochemical stability, and solubility in diverse solvents are among the unique characteristics why Nile red was selected as a solvatochromic probe in acidic DESs.[23] The spectral scan of DESs in Nile red as observed in Figure 1 shows a bathochromic shift when the

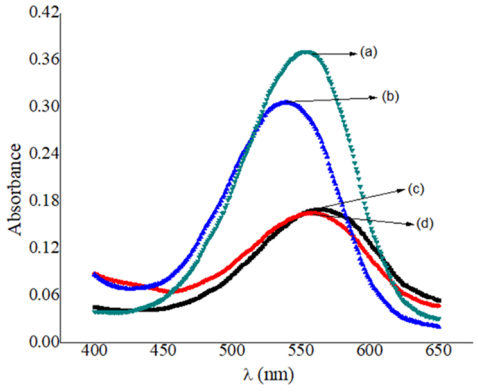


Figure 1. Absorption spectra of Nile red dye in DESs (1:3) at 298 K. (a) methanol (used as a control solvent), (b) [Ch]Cl:CA, (c) [Ch]Cl:BA, and (d) [Ch]Cl:VA. Standard uncertainty, u, of λ is $u(\lambda)$ = 0.2 nm.

hydrocarbon chain length of the hydrogen bond donor is decreased. Nile red shows a bathochromic shift with polar solvents and a hypsochromic shift in nonpolar solvents. From the results, the λ_{max} of Nile red does not shift significantly among the various DESs, specifically between the DESs formed with the same HBD at different molar ratios. The small variations in shifts of λ_{max} are related to increasing alkyl chain length of acids because of changes in the microsphere solvation of probes. Recently, Aryafard et al.[32] used molecular dynamics simulation to explain the preferential solute−solvent interaction of solvatochromic probes for different pure solvents and binary mixtures. In most cases, probes preferred to be solvated by hydrophilic sides of the solvent than their hydrophobic sides.

As shown in Figure 1, changing the alkyl chain length of the HBD impacts the polarity and the nature of the DESs synthesized. The results support a previous study that the maximum wavelength of Nile red changes marginally in the presence of hydrogen bonding solvents.[23] Most importantly, the electron transition of all the DESs synthesized stayed linear between T = (298.15 and 318.15) K except for [Ch]:BA (1:3) and [Ch]Cl:CA (1:2), which showed a modest shift in the λ_{max} between T = (308.15 and 318.15) K (Figure 2). This implies

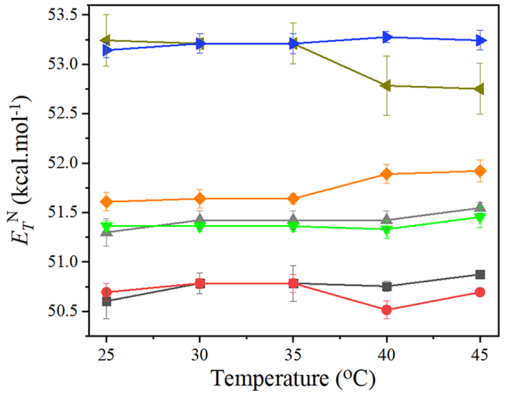


Figure 2. Electronic transition of Nile red dye in DESs and methanol (control). Black box solid = [Ch]Cl:BA (1:2), red circle solid = [Ch]Cl:BA(1:3), gray triangle up solid = [Ch]Cl:VA(1:2), bright green triangle down solid = [Ch]Cl:VA (1:3), orange diamond solid= methanol, dark yellow triangle left-pointing solid = [Ch]Cl:CA(1:2), and blue triangle right-pointing solid = [Ch]Cl:CA (1:3) DESs. E_T^N values were computed from average of triplicate readings of λ = (400 to 650) nm.

that the DES behavior changes at elevated temperatures and may affect the empirical polarity index of the solvent. The E_T (NR) can be used as a single parameter to estimate the polarity of a solvent. Additionally, the E_T (NR) value of methanol (a control solvent) at 298.15 K was found to be 51.61 ± 0.093 kcal·mol^{-1} comparable to a literature value of 52.02 kcal·mol^{-1} and Reichardt's E_T (30) and E_T (26)[23] values. Increasing the hydrocarbon chain of the HBD affects the E_T (NR) parameter and hence the polarity of the DESs. This simple and single solvent property is a good indicator for selecting appropriate DES composition for tuning the solvent especially in solvent-dependent chemical reactions.

3.2.2. Kamlet−Taft Parameters. The Kamlet−Taft parameters dipolarity/polarizability (π^*) and the hydrogen bonding basicity (β) of DESs were determined by using DENA and 4NA solvatochromic probes and employing eqs 3 and 4, respectively. The π^* measures the nonspecific interactions such as dipole−dipole and dipole-induced dipole interactions and the polarizability occurring between the dye and the DESs. From the results, nonspecific interactions serve as an index for DES polarity and it decreases with increasing hydrocarbon chain length of the HBD. For instance in Table 3, the nonspecific interactions in DESs at 313 K decrease from 0.9452 ± 0.02 of [Ch]Cl:BA (1:2) to 0.3270 ± 0.03 of

https://dx.doi.org/10.1021/acs.jced.9b00872
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Table 3. Kamlet–Taft Parameters of DESs at 313 and 318 K[a]

		π^*		β		α		E_T^N	
DESs	molar ratio	313 K	318 K	313 K	318 K	313 K	318 K	313 K	318 K
[Ch]Cl:BA	1:2	0.9452	0.9452	0.5138	0.5138	0.7751	0.7489	0.6189	0.6226
	1:3	0.9136	0.9073	0.5256	0.4990	0.8472	0.8123	0.6115	0.6171
[Ch]Cl:VA	1:2	0.8303	0.8303	0.4949	0.4775	0.7027	0.6758	0.6396	0.6434
	1:3	0.7847	0.7978	0.5572	0.5069	0.7518	0.7166	0.6367	0.6405
[Ch]Cl:CA	1:2	0.4212	0.4570	0.8527	0.8462	0.6670	0.6513	0.6816	0.6806
	1:3	0.3270	0.4069	0.9546	0.8784	0.6203	0.5765	0.6968	0.6957

[a]Average standard uncertainties, u, of T, π^*, β, α, and E_T^N are $u(T) = 0.2$ K, $u(\pi^*) = 0.011$, u(β) = 0.0068, $u(\alpha) = 0.0056$, and $u(E_T^N) = 0.0011$.

[Ch]Cl:CA (1:3), signifying that a slight change in the donor group significantly alters the π^* parameter.

Also, changing the temperature to 318 K gave a similar trend indicating that hydrogen bond donors contribute significantly to the nonspecific interactions and polarity of the solvent. These results confirm a recent report by Teles et al.[33] that π^* values decrease by increasing alkyl chain length between ammonium-based salts and carboxylic acids.[23] The ionic character of [Ch]Cl:BA (1:2) DES is the highest among the prepared DESs probably due to the presence of permanent dipoles and delocalized bonds[28] in the solvent, which might give rise to higher π^* values. The dipolarity/polarizability values of methanol (control) at 313 K are 0.76495 ± 0.030 and 0.82378 ± 0.030 at 318 K, which are closer to [Ch]Cl:VA (1:3) and [Ch]Cl:VA (1:2) DESs, respectively.

The β describes the ability of a solvent to accept protons or donate electron density to form hydrogen bonding and is mostly affected by the nature of the anionic part of the HBA.[33] From the results, as opposed to π^*, which decreases by increasing the hydrocarbon chain of the HBD, a reverse trend is observed in the β parameter. Apart from [Ch]Cl:BA (1:2) DES, which showed no change in β, a modest decrease was observed in DESs at T = (313 to 318) K (Table 3). As described,[34] DES-ethaline, DES-glyceline, and DES-reline data show that β and π^* parameters remain constant with increasing temperature. In contrast to these earlier reports, π^* and β values of ionic liquids [bmim][PF_6] and [bmim]-[BF_4], respectively, were found to decrease with increasing temperature.[35] In this study, the π^* values increase marginally between T = (298 and 318) K when BA and VA are used as HBD (Figure 3). Unlike other reports, our data was collected within a short range of temperature compared to literature values collected at T = (298.15 to 368.15) K and may account for the deviation. The β values of the DESs show their ability to accept protons for chemical reactions, which are even better than some organic solvents or ionic liquids. The Kamlet–Taft hydrogen bonding acidity (α) describes the ability of the solvent to donate protons and is calculated from the E_T (NR) and π^* values (eqs 1 and 2). From the results, the hydrogen bonding acidity of DESs ranges from 0.620280 to 0.847230 at 313 K. The high acidic nature of the DESs may be due to the carboxylic functional group present in the DESs. Increasing the alkyl chain or the molar ratio of the HBD has significant impact on the α parameter.

To elucidate the polarity of the synthesized DESs based on calculated α, β, and π^* values, a ternary plot was created to compare our data with some selected organic solvents and ionic liquids. The ternary plot helps to show the polarity relationship of DESs, ionic liquids, and organic solvents based on the Kamlet–Taft polarity indices. In Figure 4, the α, β, and

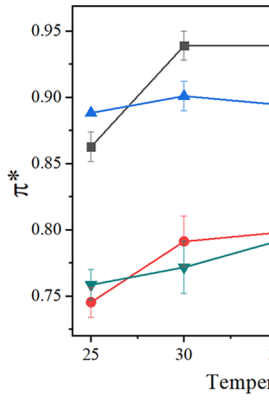


Figure 3. Dipolarity/Polarizability of DESs at different temperatures. Black box solid = [Ch]Cl:BA (1:2), red circle solid = [Ch]-Cl:VA(1:2), blue triangle up solid = [Ch]Cl:BA (1:3), and teal triangle down solid = [Ch]Cl:VA (1:3).

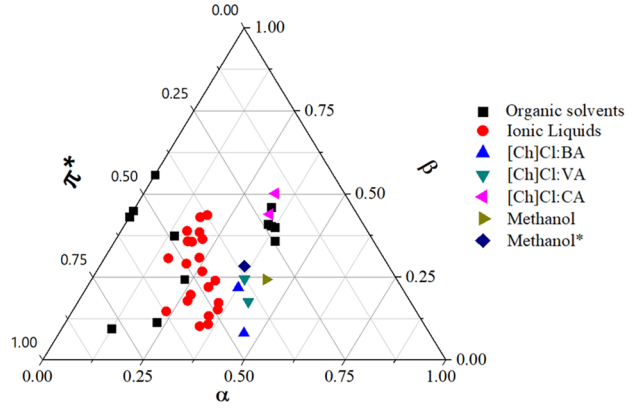


Figure 4. Kamlet–Taft parameters of newly synthesized DESs at 298 K compared to selected organic liquids and ionic liquids. The references for α, β, and π^* values for ionic liquids and organic solvents are listed in Table S4. *Control for this work.

π^* values of [Ch]Cl:BA (1:2) and [Ch]Cl:BA (1:3) are within the range of ionic liquids, whereas [Ch]Cl:VA (1:2) and [Ch]Cl:VA (1:3) are in the close range to Kamlet–Taft parameters of methanol. The Kamlet–Taft parameters of the ethanol, 1-propanol, 1-butanol, and *t*-butanol cluster closer to [Ch]Cl:CA DES, which depict similarity of their polarity indices. These results were validated because the experimental α, β, and π^* values of methanol (that is, methanol* on the ternary plot) obtained are similar to α, β, and π^* values of methanol obtained from the literature (Table S4). With respect to altering the hydrogen bond donor, the Kamlet–Taft

https://dx.doi.org/10.1021/acs.jced.9b00872
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parameters that appeared more susceptible to change are hydrogen bonding acidity and dipolarity/polarizability.

3.3. Density Functional Theory. *3.3.1. Bond Angle and Bond Distance.* The bond angle and bond distance of DESs were obtained from the optimized geometry of HBA and HBD molecules through DFT/B3LYP calculation with the 6-31G (d) basis set. DES molar ratios (acceptor:donor) used for the study are 1:1, 1:2, and 1:3 except for [Ch]Cl:CA, which 1:3 was omitted due to computational time. The optimized geometries, bond distance, and bond angles of [Ch]Cl:BA (1:2) are shown in Figure 5. The bond angles, bond distances,

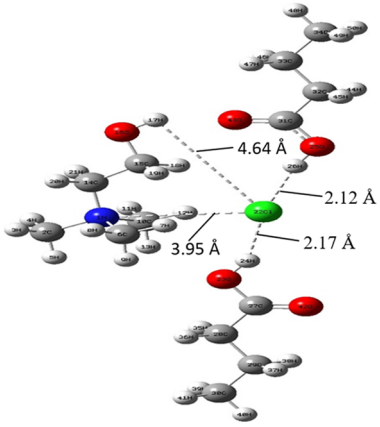


Figure 5. Optimized geometry of [Ch]Cl:BA (1:2) DES computed by DFT/B3LYP with 6-31G (d) basis set. Atoms are colored according to type: C (gray), O (red), N (blue), H (white), and Cl (light green). Bond angle between C14−C15−O16 (104.44°), C15−O16−H17 (104.09°), C2−N1−C14 (107.99°), and N1−C14−C15 (115.53°).

and optimized geometries of the remaining DESs are found in the Supporting Information. To study the effect of the hydrogen bond donor on the choline chloride acceptor molecule, we compared the bond angles of the atoms in the simulated DESs and the optimized acceptor. The bond angles for C15−O16−H17 and N1−C14−C15 in the optimized choline chloride are 117.34° and 106.96°, respectively. These angles are more susceptible to change if molecular interactions occur between the acceptor and the hydrogen bond donor. Moreover, the degree of such interaction can be quantified by measuring the bond angles. The [Ch]Cl:BA, [Ch]Cl:VA, and [Ch]Cl:CA DES (1:2) showed a reduced bond angles of 115.53°, 116.09 °, and 116.09 °, respectively, for C15−O16−H17, and 104.09°, 105.01°, and 105.02°, respectively, for N1−C14−C15. The lowering of bond angles for both C15−O16−H17 and N1−C14−C15 in DESs shows that molecular interaction occurs between the HBA and the HBD during DES formulation. Also, the calculated bond distance between the carboxylic proton of the HBD and the chloride anion of the HBA in the optimized DESs was used to verify hydrogen bonding. All the carboxylic protons are within a bond distance of $2.04 \leq$ Å ≤ 2.22, which can possibly form a short-strong hydrogen bonding. This interaction may be responsible for the eutectic nature of the formulated mixture.

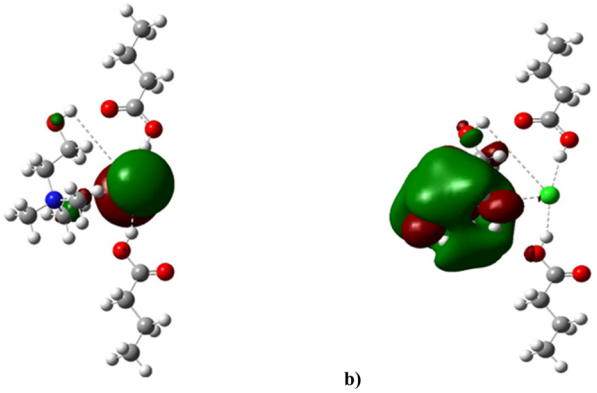
3.3.2. HOMO and LUMO Energies. The highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO) were calculated for DESs including their corresponding constituents. These orbitals are important in quantum chemistry because electrons are mostly located in them for reactions. The HOMO defines a region where electrons have the highest energy for reaction, whereas the LUMO creates a region for invading electrons from other molecules to fill-in during bonding. In other terms, the HOMO donates electrons (ionization potential) into the LUMO, which tend to act as the orbital's electron affinity (ability to obtain electrons). From the results (Table S2), the ionization potential of the HBD decreases as the alkyl increases. As expected, the ionization potential of the HBD is greater than the HBA and the electron affinity of the HBA is greater than the HBD. Among the hydrogen bond donors, those with two molar ratios have the highest ionization potential (Table S2).

The difference in energy between these frontier molecular orbitals (HOMO−LUMO gap) is critical to study molecular reactivity, global hardness, electronegativity, chemical potential, and charge transfer within the molecule. The higher the HOMO−LUMO gap, the less reactive the molecule because electrons have wider distance to jump and the molecule becomes more stable. Among the raw materials studied, choline chloride has the smallest electronic gap of 0.18477 eV, which makes it most reactive and least stable compared to all the HBD with electronic gaps between 0.2688 and 0.28527 eV. The global hardness (wider HOMO−LUMO gap) of the DESs increased marginally when the molar ratio of the HBD changed but no apparent change occurred by increasing the alkyl chain length. The hardness and softness (shorter HOMO−LUMO gap) property of DESs is important especially when the solvent is applied in chemical reactions, organic synthesis, pharmaceuticals, and electrochemistry. The clear understanding of this phenomenon is essential to help choose the correct solvent for desired industrial and pharmaceutical purposes.

3.3.3. Charge Transfer (ΔN). As stated by Sanderson in the electronegativity equalization principle,[36,37] the combination of HBA and HBD leads to electron flow from the species of lower electronegativity to species of higher electronegativity until the adduct (in this case, DESs) electronegativity becomes equalized. The electronegativity of the HBA, HBD, and DESs were calculated from the FMOs using Mulliken's formulae[37,38] [$\chi = (I + A)/2$]. From the results (Table S2), electrons flow from the HBA ($\chi = 0.079475$) to the HBD ($0.131845 \leq \chi \leq 0.13823$). The electronegativity values of the DESs are closer to the geometric mean of the starting raw materials. The chemical potential (Lagrange multiplier, $\mu = -\chi$), a new density functional parameter by Parr et al.,[17,37] was calculated as an index to measure charge transfer between the HBA and the HBD in the formulated DESs. The electronegativity and chemical potential of DESs are similar when the same molar ratio of HBD is used irrespective of the alkyl chain length (Table S3). As shown in Figure 6, the electron flow pattern indicates that the HOMO structure of [Ch]Cl:BA (1:2) has electron density localized on the chloride anion (light green) while the LUMO electron cloud is localized on the nitrogen (blue) and hydrogen (white) of the HBA. As displayed in Table 4, the HOMO and LUMO data obtained by the DFT calculations were used to calculate charge transfer occurring within the solvent. Also, the calculated electronic gap is a measure of the DES stability.

As shown in Table 5, it is not surprising that the [Ch]Cl:BA (1:2) DES has the highest charge transfer of 0.250806 followed by 0.248927 of [Ch]Cl:VA (1:2) and 0.246531 of [Ch]Cl:CA (1:2). Abbott et al. showed that charge transport influences DES viscosity by controlling ionic mobility.[2] Interestingly, the high charge transfer values obtained by DFT calculations of DESs at 1:2 molar ratio confirm the necessity of two carboxylic

https://dx.doi.org/10.1021/acs.jced.9b00872
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a) b)

Figure 6. Frontier molecular orbitals of [Ch]Cl:BA (1:2) DES. (a) HOMO and (b) LUMO atoms are colored according to type: C (gray), O (red), N (blue), H (white), and Cl (light green).

Table 4. Electronic Properties of Carboxylic Acid-Based DESs[a]

DESs	molar ratio	ϵ HOMO (eV)	ϵ LUMO (eV)	$\Delta\epsilon$ = electronic gap (eV)
[Ch]Cl:BA	1:1	−0.20929	0.00919	0.21848
	1:2	−0.21985	0.01153	0.23138
	1:3	−0.23722	0.01036	0.24758
[Ch]Cl:VA	1:1	−0.20937	0.00909	0.21846
	1:2	−0.22225	0.01279	0.23504
	1:3	−0.2372	0.01044	0.24764
[Ch]Cl:CA	1:1	−0.20855	0.01129	0.21984
	1:2	−0.22162	0.01288	0.2345

[a]Standard uncertainties, u, of HOMO, LUMO, and ϵ are u(HOMO) = 0.014 eV, u(LUMO) = 0.00083 eV, and $u(\epsilon)$ = 0.015 eV.

Table 5. Molecular Indices of DESs by Density Functional Theory[a,b]

DESs	mole ratio	Debye	ΔN
[Ch]Cl:BA	1:1	10.2790	0.224412
	1:2	11.6059	0.250806
	1:3	10.6744	0.239059
[Ch]Cl:VA	1:1	10.3810	0.223714
	1:2	11.9701	0.248927
	1:3	10.6264	0.237014
[Ch]Cl:CA	1:1	9.36920	0.222832
	1:2	11.9343	0.246531

[a]The charge transfer is calculated by the formulae:[17] $\Delta N = \mu B - \mu A/\eta B + \eta A$. [b]Standard uncertainty, u, of ΔN, $u(\Delta N)$ = 0.014.

acids for DES formation. Moreover, Ghosh and Bhattacharyya[17] used density functional theory to find the charge transfer of the $F_3B{-}NH_3$ covalent adduct as 0.264178. The lower charge transfer in DESs unlike the conventional covalent bonding in $F_3B{-}NH_3$ could explain the weaker interaction between HBA and HBD in the eutectic mixture.

Apart from charge transfer, which serves as an index of solvent polarity, dipole moments (the chief quantitative descriptor of molecular charge distribution[36]) of all the molecules were calculated. Dipole moments of the hydrogen bond donor, hydrogen bond acceptor, and DESs show that the molecules undergo conformational reorientation during DES formation. Also, the dipole moments increase to approximately 12 Debye in [Ch]Cl:BA, [Ch]Cl:VA, and [Ch]Cl:CA DESs at 1:2 molar ratio. The DFT results presented herein could explain why the pioneering work by Abbott et al.,[2] which speculated that two carboxylic acids are required to form some carboxylic acid-based DESs.

4. CONCLUSIONS

A modest attempt to study physicochemical and thermophysical properties of carboxylic acid-based DESs by using solvatochromic probes and density functional theory was studied. A bathochromic shift was observed when [Ch]Cl:BA (1:3) DES interacts with a Nile red solvatochromic probe. An increase in the alkyl chain length of the hydrogen bond donor decreases the polarity of the DESs. Moreover, the electronic transition of DESs in Nile red is a single parameter, which can be used to estimate the polarity of the formulated solvent. Also, the Kamlet−Taft hydrogen bonding acidity (α), hydrogen bonding basicity (β), and polarizability/dipolarity (π^*) of the newly formed DESs were comparable to selected ionic liquids and organic solvents in the literature. In addition, density functional theory expanded our understanding that DESs at 1:2 molar ratio have high charge transfer and dipole moments occurring within the solvent. The DFT results confirm earlier assertions that two carboxylic acids are required for some DES formation.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jced.9b00872.

Optimized equilibrium geometry of carboxylic acid-based deep eutectic solvents, viscosity of DESs synthesized, molecular indices of DESs, and Kamlet−Taft parameters used for the ternary plot (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors are grateful to Dr. Mohammad A. Halim and Md Sajjadur Rahman of Red-Green Research Centre, Bangladesh, for the computational software.

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